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VALUING BMW'S DATA

CONFIDENTIAL



CLEMSON
CREATIVE INQUIRY +
UNDERGRADUATE RESEARCH



The Problem Space

Two intertwined challenges demand a unified answer.

Security & Exposure

Third-party AI pipelines can extract proprietary information from your video and audio.

- Sensitive speech picked up by AI microphones
- Proprietary processes and layouts visible on camera
- No visibility into how vendors use or retain your data

Data Quality & Value

Feeding AI on the wrong data is as dangerous as feeding it no data. What you build on defines your output.

- Which data attributes drive meaningful AI performance?
- Of the data points available, which carry the most predictive value?
- How do we quantify that value before signing a vendor contract?

Both problems demand the same answer: find the value of BMW's data.

Risk Scoring Framework

Our model converts BMW's qualitative risk judgment into a consistent, explainable scoring system — preserving human decision-making.

1



ASSIGN SIGNAL SCORES (s_i)
FROM VSS OUTPUT

2



APPLY ATTRIBUTE WEIGHTS (w_i)

3



CALCULATE WEIGHTED CONTRIBUTIONS

4



SUM ALL CONTRIBUTIONS

5



CONVERT TO RISK SCORE

Scoring Model

$$\text{Risk Score} = 100 \times \sum w_i \cdot s_i$$

$i = 1, 2, \dots, 6$ (6 risk attributes)

w_i = attribute weight

s_i = signal strength (0–1)

Output = 0–100 → ICL level

0 – 25



NOT PUBLIC

Basic Protection

DAMAGE POTENTIAL GUIDANCE

Up to approximately
€10,000

26 – 45



CONFIDENTIAL

Basic Protection

DAMAGE POTENTIAL GUIDANCE

Up to approximately
€1 Million

46 – 70



STRICTLY CONFIDENTIAL

Special Protection

DAMAGE POTENTIAL GUIDANCE

Up to approximately
€100 Million

71 – 100



SECRET

Highest Protection

DAMAGE POTENTIAL GUIDANCE

Over **€100 Million**

Key Words & Attributes

SIGNAL SCORE → ● 0.0 No signal ● 0.2 Not public ● 0.5 Confidential ● 0.8 Strictly confidential ● 1.0 Secret

w = 0.25 Process Transparency

- **NOT PUBLIC** Walking through, passing by, background activity
- **CONFIDENTIAL** Stacking boxes, scanning packages, routine task
- **STRICTLY CONF.** "then," "next," "followed by" — sequential steps visible
- **SECRET** Complete assembly sequence, full workflow exposed

w = 0.20 Equipment Visibility

- **NOT PUBLIC** Shelf, rack, bin, cart, pallet — generic storage
- **CONFIDENTIAL** Forklift, conveyor belt, standard press, hydraulic lift
- **STRICTLY CONF.** Robotic arm, CNC, AGV, PLC panel, laser scanner
- **SECRET** BMW-specific tooling, proprietary fixture, custom jig

w = 0.15 Process Stage Sensitivity

- **NOT PUBLIC** Warehouse, loading dock, shipping, storage area
- **CONFIDENTIAL** General production floor, machining area, subassembly
- **STRICTLY CONF.** Paint booth, body shop, BIW, stamping, clean room
- **SECRET** Final assembly, powertrain, battery pack, prototype line

w = 0.15 Proprietary Method Exposure

- **NOT PUBLIC** Worker present, standing near equipment, idle
- **CONFIDENTIAL** Measuring tape, clipboard, visual inspection, basic tools
- **STRICTLY CONF.** Custom gauge, proprietary fixture, non-standard process
- **SECRET** BMW-specific calibration, classified work method

w = 0.15 Human Exposure

- **NOT PUBLIC** Single worker, no group activity, operating independently
- **CONFIDENTIAL** Two-four workers, nearby, same aisle or work cell
- **STRICTLY CONF.** Large group, side by side, jointly reviewing same item
- **SECRET** Hands-on demonstration, direct method instruction

w = 0.10 Sensitive Data Visibility

- **NOT PUBLIC** Screen in background, document present but unreadable
- **CONFIDENTIAL** Barcode, part number, work order, standard form visible
- **STRICTLY CONF.** Process data on screen, ID badge, SOP, KPI dashboard
- **SECRET** BMW internal system, classified document, restricted data

! Flag 1 · Third-Party Presence

visitor badge · person being escorted · contractor observing · external consultant · supplier on floor · non-BMW personnel · guided tour in progress · audit team reviewing · media in facility

! Flag 2 · Critical Data Detection

customer record · client data visible · employee record · personnel file · HR data · contract visible · legal document · contractual terms · compliance data

Risk Scoring Framework Application

Purpose: Convert VSS text into structured, transparent indicators that support — never replace — BMW risk team judgement

Framework Overview

Structured scoring pipeline converting VSS observations into a confidentiality risk assessment.

- 6 Confidentiality Attributes
- Simple Scoring Equation
- ICL Mapping
- Maximum Principle
- Escalation Flags
- Structured Report



How it Helps the Risk Team

Elevating analyst judgment through structured, transparent, and auditable confidentiality scoring.

- Improves Consistency
- Sharpens Analysts Focus
- Enhances Decision Quality
- Builds Transparency
- Keeps Humans in Control

Outcome: Better, faster, and more defensible Confidentiality decisions through automation that elevates, not replaces, expert judgement

Hedonic Dimensional Analysis

The valuation approach we researched — and how it works with VSS data.

THE CORE IDEA

A dataset's total value is the sum of its underlying attributes.

Instead of attempting to value a raw video feed, we decompose the dataset into observable attributes (our dimensions) and aggregate their weighted scores to determine overall strategic value.

HOW IT APPLIES TO VSS

Keywords as observed attributes

We use keyword frequency to identify and measure these attributes. When a camera log repeatedly records high-value language (e.g., 'robotic arm' or 'stoppage'), it provides a measurable signal that a specific dimension is strongly represented in the data.

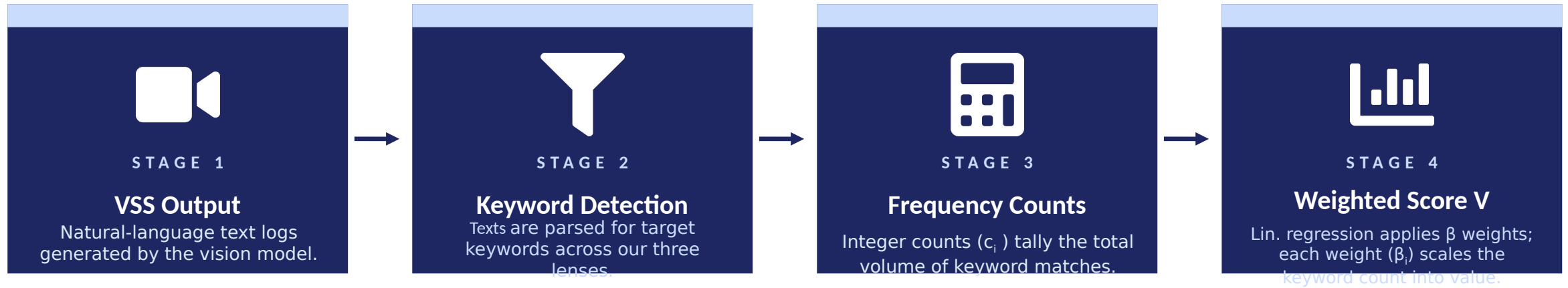
WHAT IT CLAIMS

Correlation, not causation

We are not claiming that specific keywords independently generate monetary value. Rather, we are measuring correlation: as the frequency of operationally rich language increases in the logs, the analytical utility of the dataset rises proportionally.

Keywords → VSS → Score

The pipeline from raw factory video to a weighted dataset value.



HOW KEYWORDS WERE DETERMINED

ADD VALUE

Standard VSS terms that indicate a dimension is observable. Example (1.1 Process ID): welding, torquing, assembling, painting, stamping.

SUBTRACT / FLAG

Uncertainty language or disqualifying content. Example (3.4 Confidence): "appears to," "may be," "unclear," "approximately."